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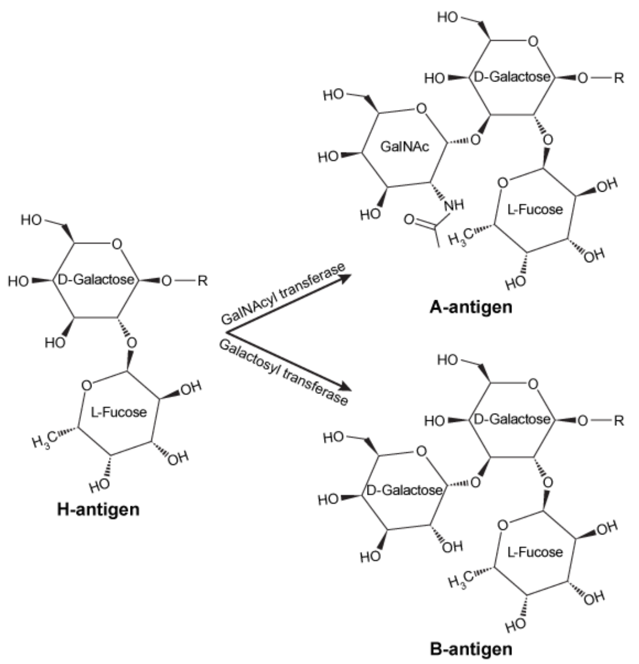

Figure 1 Structures of the H-, A-, and B-antigens. The R-group represents additional carbohydrates, which are linked to galactose through a glycosidic bond.

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GL6	1019.82	20:1

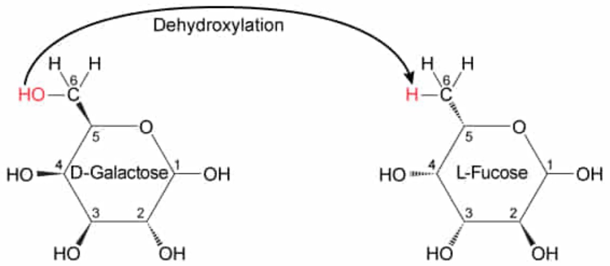
Ignoring the anomeric carbon, the fucose shown in Figure 1 is the dehydroxylated enantiomer of:

- A. L-xylulose. [8%]
- B. D-ribulose. [21%]
- C. L-glucose. [27%]
- ✔ D. D-galactose. [42%]

Correct
42% answered correctly

Explanation:

L-fucose is the dehydroxylated enantiomer of D-galactose



All stereocenters are in opposite configurations

Note: Configuration of C1 (anomeric carbon) is not shown because it can adopt either configuration

Enantiomers are molecules with the **same molecular formula** that are mirror images because they differ in the orientation of every stereocenter. In planar diagrams of sugars, the orientation of stereocenters is generally indicated by bonds that extend above (wedge bonds) or below (dash bonds) the plane of the rings. The enantiomer of a D-sugar is the L-form of the same sugar. For example, the enantiomer of D-mannose is L-mannose. Most biologically relevant sugars adopt the D-configuration instead of the L-configuration, with fucose being a rare exception.

Figure 1 shows the structure of L-fucose (not to be confused with fructose) as part of the H-antigen. L-fucose is dehydroxylated at **carbon 6** and is connected to its hydroxyl substituents by **dash bonds** at carbons 3, 4, and 5 and by **wedge bonds** at carbon 2. The question says to ignore the wedged anomeric carbon (carbon 1), and carbon 6 is achiral, so the correct sugar only needs to have opposite stereochemistry from L-fucose at carbons 2, 3, 4, and 5. The reason the anomeric carbon can be ignored is that monosaccharides can adopt either configuration through mutarotation.

As shown in Figure 1, **D-galactose** contains a hydroxyl group at carbon 6, wedge bonds at carbons 3, 4, and 5, and a dash bond at carbon 2. Therefore, when D-galactose is superimposed on top of L-fucose, it becomes apparent that the two molecules have opposite stereochemistry at every stereocenter (excluding carbon 1, the anomeric carbon). For this reason, **L-fucose** is also known as 6-deoxy-L-galactose and is the **dehydroxylated enantiomer** of D-galactose.

(Choices A and B) Xylulose and ribulose are pentoses (five-carbon sugars) formed during the pentose phosphate pathway. L-Fucose is a dehydroxylated hexose (six carbons), so it cannot be the enantiomer of a pentose. In addition, L-xylulose can only be the enantiomer of a D-sugar.

(Choice C) The fucose shown in Figure 1 is in the L-configuration, so an L-sugar such as L-glucose cannot be the enantiomer.

Educational objective:
Enantiomers are molecules with identical molecular formulae that differ in the configuration of every stereocenter. They are mirror images of each other. The enantiomer of a D-sugar is the L-form of the same sugar.

Time spent:0 sec	**QID**:402317
Subject: Biochemistry	**System:** Carbohydrates, Nucleotides, and Lipids

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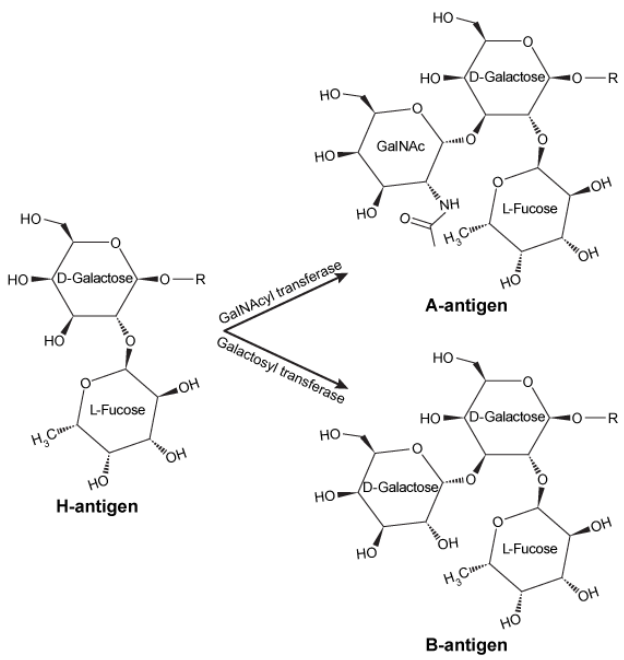

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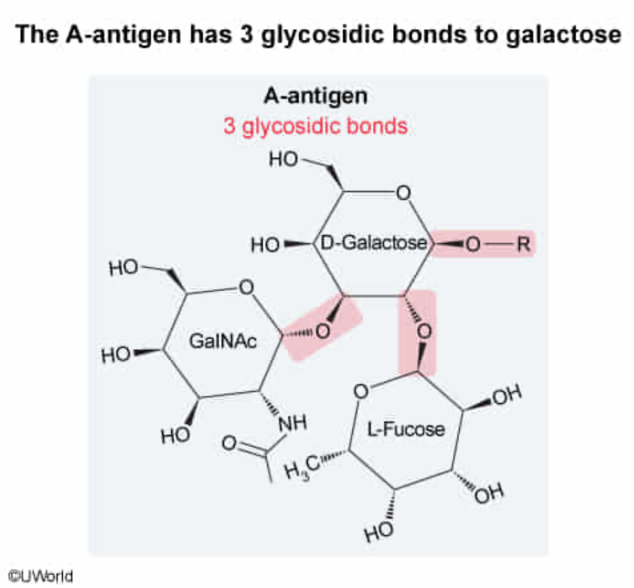
The D-galactose moiety of the A-antigen participates in how many glycosidic bonds total?

- A. 1 [4%]
- B. 2 [49%]
- ✔ C. 3 [45%]
- D. 4 [1%]

Correct
44% answered correctly

Explanation:



A glycosidic bond is a bond between the anomeric carbon (ie, the hemiacetal or hemiketal) of a carbohydrate and **any other biological molecule**, including proteins, lipids, nucleotides, and other carbohydrates. A **single carbohydrate** can participate in **multiple glycosidic bonds**: one through its own anomeric carbon and others through bonds from its hydroxyl groups *to* the anomeric carbons of *other* sugars. The various ways in which carbohydrates can link to each other give rise to a high level of diversity among carbohydrate chains.

Figure 1 shows that the D-galactose moiety of the A-antigen participates in **three glycosidic bonds**: One linking the anomeric carbon of D-galactose to the R-group (which the passage states is a polysaccharide), one linking the anomeric carbon of L-fucose to carbon 2 of D-galactose, and one between the anomeric carbon of GalNAc and carbon 3 of D-galactose.

(Choice A) The terminal sugars GalNAc and L-fucose each participate in 1 glycosidic bond, but D-galactose participates in 3.

(Choice B) *In the H-antigen,* the D-galactose moiety participates in 2 glycosidic bonds; however, note that another glycosidic bond is added to D-galactose in the A-antigen, which the question specifies.

(Choice D) D-galactose would require another glycosidic bond at carbon 4 or carbon 6 to have a total of 4 glycosidic bonds.

Educational objective:

A glycosidic bond is a bond between the anomeric carbon of a carbohydrate and any other biomolecule. A single carbohydrate may participate in many glycosidic bonds by linking to the anomeric carbons of other carbohydrates, allowing a high level of diversity among carbohydrate chains.

Time spent: 0 sec	**QID**:402318
Subject: Biochemistry	**System:** Carbohydrates, Nucleotides, and Lipids

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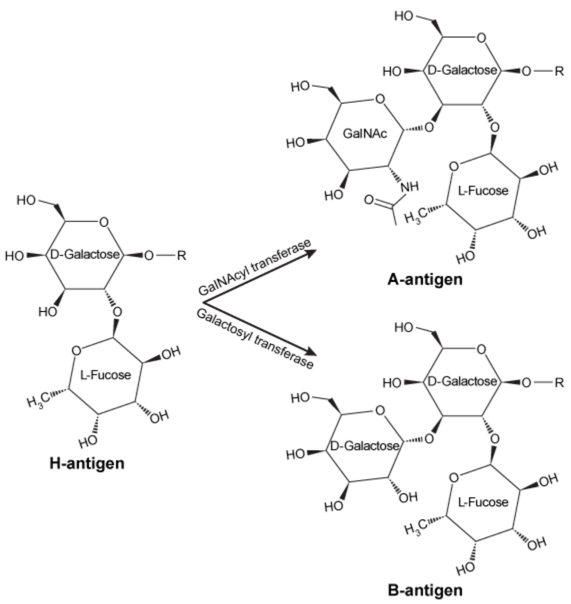

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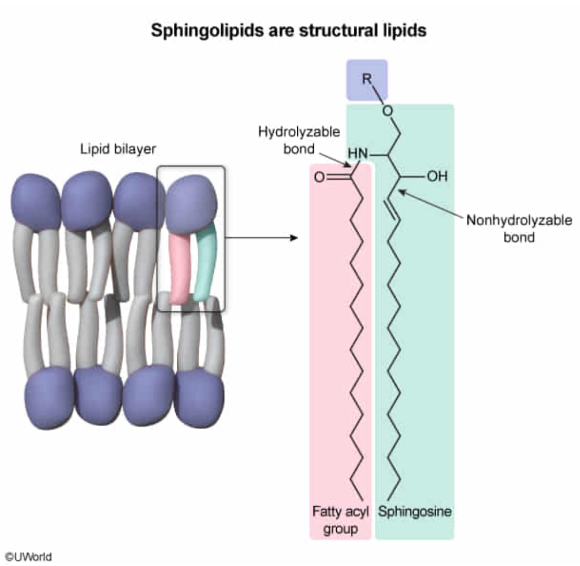
Which of the following statements accurately describes the sphingolipids in Table 3?

I. They provide structure to biological membranes
II. They are primarily used for energy storage
III. They can be hydrolyzed to produce two fatty acids

- ✔ ◉ A. I only [44%]
- ○ B. III only [18%]
- ○ C. I and II only [21%]
- ○ D. II and III only [16%]

Correct
44% answered correctly

Explanation:



Lipids are broadly classified as either hydrolyzable or nonhydrolyzable. They can also be grouped by function as **structural lipids**, **energy-storing lipids**, and **signaling lipids**. Structural lipids are found in the membranes surrounding cells and organelles and include hydrolyzable glycerophospholipids and sphingolipids as well as nonhydrolyzable cholesterol. The properties of individual lipids such as the number of **fatty acid tails**, **double bonds**, and head group structures influence **both fluidity and curvature** of biological membranes. The sphingolipids in Table 3 contain a sphingosine backbone attached to a single fatty acyl chain and a polar head group. In red blood cells and other cell types these sphingolipids are essential structural components of the plasma membrane **(Number I)**.

(Number II) The lipids that are primarily responsible for energy storage are triacylglycerides, which are metabolized through hydrolysis and beta-oxidation. The fatty acyl groups of sphingolipids may be hydrolyzed and used for energy production, but this is not their primary function.

(Number III) Glycerophospholipids contain two hydrolyzable fatty acyl chains, but sphingolipids contain only one. The other long chain hydrocarbon in a sphingolipid is part of the sphingosine backbone and cannot be hydrolyzed. Therefore, sphingolipids produce *one* fatty acid upon hydrolysis, not two.

Educational objective:
Sphingolipids are structural lipids that help influence the fluidity and curvature of biological membranes. The long hydrocarbon chain of the sphingosine head group cannot be readily hydrolyzed, and sphingolipids are not a primary means of energy storage.

Time spent:0 sec	**QID**:402319
Subject: Biochemistry	**System:** Carbohydrates, Nucleotides, and Lipids

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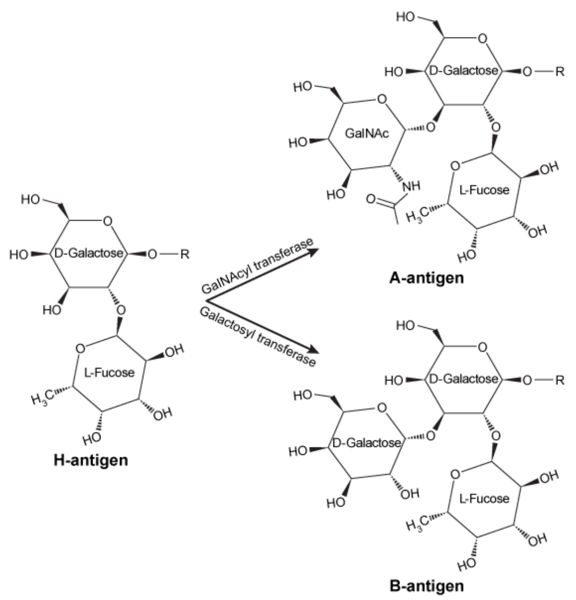

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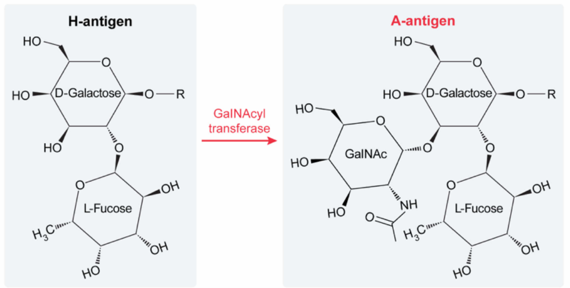
GL6	1019.82	20:1

Approximately what percentage of the population measured in Table 2 expresses GalNAcyl transferase?

- A. 29% [72%]
- B. 27% [20%]
- C. 11% [4%]
- D. 9% [2%]

Correct
73% answered correctly

Explanation:



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2920/10,000 ≈ 29%

According to the passage, **GalNAcyl transferase** modifies the H-antigen by adding GalNAc in an α-1,3 glycosidic linkage to D-galactose, producing the **A-antigen**. Therefore, individuals who express the A-antigen on their red blood cells must express at least one copy of GalNAcyl transferase. This includes participants in the study with **type A blood** and those with **type AB blood**.

Table 2 shows that 2,746 participants had type A blood and 174 had type AB blood, for a total of 2,920 participants who expressed the A-antigen. Therefore, 2,920 participants expressed GalNAcyl transferase. The total number of participants was 10,000, so **approximately 29%** of the participants in the study expressed GalNAcyl transferase.

(Choice B) Approximately 27% of participants had type A blood (2,746/10,000), but individuals with type AB blood also express GalNAcyl transferase and must be counted in the calculation.

(Choice C) Those with type B blood do *not* express GalNAcyl transferase. Approximately 11% of the participants had either type B blood or type AB blood (887+174/10,000). This represents the percentage of individuals who express *galactosyl* transferase.

(Choice D) Approximately 9% of participants had type B blood (887/10,000). These individuals express *galactosyl* transferase, not GalNAcyl transferase.

Educational objective:

To determine the percentage of a population that meet specific criteria, all individuals with the relevant trait must be considered out of the total population evaluated.

Time spent:0 sec
Subject:
Biochemistry

QID:402320
System:
Carbohydrates, Nucleotides, and Lipids

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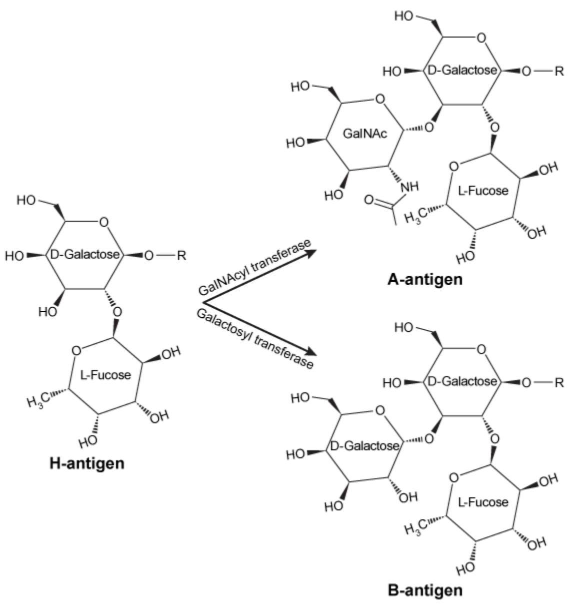

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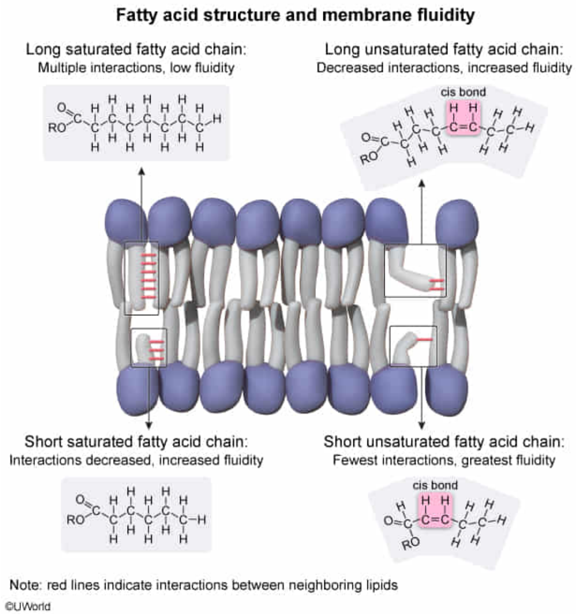
Which lipid in Table 3 will cause the greatest increase in fluidity if inserted into a membrane?

- A. GL2 [12%]
- B. GL3 [10%]
- C. GL5 [18%]
- ✔ D. GL6 [58%]

Correct
59% answered correctly

Explanation:



The **fluidity and rigidity** of plasma membranes is partially dependent on the **structure of fatty acyl tails** in the lipid bilayers. **Increasing intermolecular interactions** between fatty acyl tails leads to decreased motion and **more rigidity** (less fluidity). The **length** and **number of double bonds** in a fatty acyl tail influence the number of hydrophobic interactions possible.

Short tails participate in **fewer** intermolecular interactions than long tails and therefore result in **increased fluidity**. Carbon-carbon double bonds in fatty acyl tails are typically in the **cis-conformation** and introduce a bend in the tail, **preventing** adjacent tails **from interacting efficiently** with each other. A **higher number of double bonds** leads to decreased interaction and **greater fluidity**.

The lipid in Table 3 that has the shortest fatty acyl tail length and the most double bonds will most likely cause the greatest increase in membrane fluidity if inserted into the membrane. GL6 and GL3 have the shortest fatty acyl tails (20 carbons), but GL6 has more double bonds (one instead of zero). Therefore, **GL6 most likely causes the greatest increase** in membrane fluidity.

(Choice A) GL2 has the longest tail of the lipids in the table and has zero C=C double bonds. It causes the *smallest* increase in membrane fluidity. In fact, it is likely to make the membrane *less* fluid (more rigid).

(Choice B) GL3 and GL6 both have the shortest tail lengths, but GL3 has no C=C double bonds, so it induces a smaller increase in membrane fluidity than GL6.

(Choice C) GL5 has a double bond, but it has a longer tail than GL6, so GL5 causes a smaller increase in membrane fluidity.

Educational objective:
The fluidity of a cell membrane is largely dependent on the lengths of the fatty acyl tails in the bilayer and on the number of double bonds in each tail. Short chains with double bonds yield the highest fluidity because they participate in the fewest intermolecular

interactions with neighboring lipids.

Time spent: 0 sec	**QID**:402321
Subject:	**System:**
Biochemistry	Carbohydrates, Nucleotides, and Lipids

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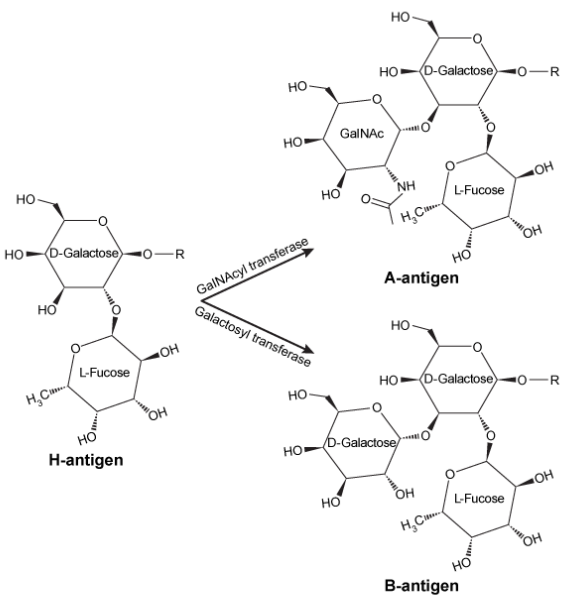

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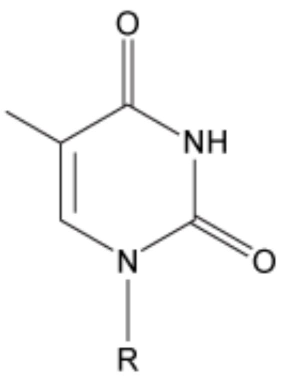
GL3	1020.87	20:0
GL4	1034.36	22:1
GL5	1048.50	24:1
GL6	1019.82	20:1

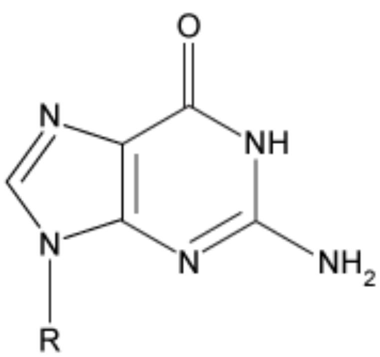
An individual with type B blood would most likely have which nucleotide at position 793 of the ABO gene?

A.



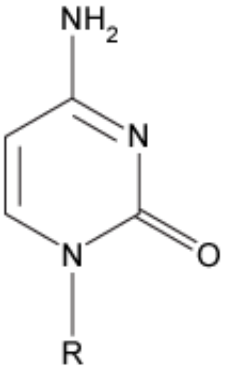
[4%]

B.



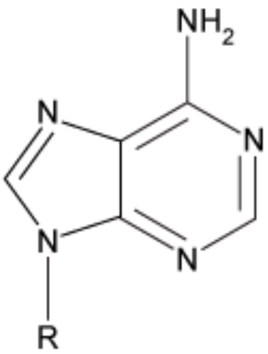
[13%]

○ C.



[7%]

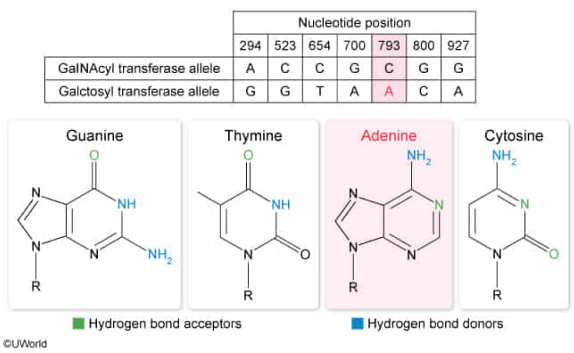
✔ ◉ D.



[74%]

Correct
74% answered correctly

Explanation:



	Nucleotide position						
	294	523	654	700	793	800	927
GalNAcyl transferase allele	A	C	C	G	C	G	G
Galctosyl transferase allele	G	G	T	A	A	C	A

The four nucleotides found in DNA (adenine, cytosine, guanine, and thymine) can be **identified by their ring structures** and by the number of **Watson-Crick hydrogen bond donors and acceptors** on each. The **purines** (adenine and guanine) each have **two rings** whereas the **pyrimidines** (cytosine and thymine) each have **one ring**.

Hydrogen bond donors can be either a nitrogen or an oxygen atom with hydrogen attached. Acceptors can take the form of oxygen or nitrogen with a lone electron pair. **Guanine and cytosine** each have **three hydrogen bond participants**: An acceptor and two donors in guanine, and a donor and two acceptors in cytosine. **Adenine and thymine** have **one donor and one acceptor each** involved in base pairing.

Figure 1 shows that the B-antigen is produced by galactosyl transferase. Table 1 shows that the galactosyl transferase allele has **adenine (A) at position 793**. Adenine is represented in **Choice D**, which shows a purine with one hydrogen bond donor and one acceptor.

(Choice A) This option shows thymine, which is not present at position 793 in either allele of the ABO gene.

(Choice B) This option shows guanine which, like adenine, is a purine. Unlike adenine, guanine has one hydrogen bond acceptor and two donors.

(Choice C) This option shows cytosine, which is the nucleotide at position 793 for the GalNAcyl transferase allele. This allele corresponds to type A blood (or type AB), but not type B blood.

Educational objective:
Nucleotide bases can be identified by their ring structure (purine or pyrimidine) and by the number of Watson-Crick hydrogen bond donors and acceptors. Adenine and guanine are purines, whereas thymine and cytosine are pyrimidines. Thymine and adenine have one donor and one acceptor each, whereas guanine has an acceptor and two donors and cytosine has two acceptors and one donor.

Time spent:0 sec
Subject:
Biochemistry

QID:402322
System:
Carbohydrates, Nucleotides, and Lipids